

Yasmine

Senior Data Scientist & Machine Learning Engineer

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PROFESSIONAL SUMMARY

Senior Data Scientist and Machine Learning Engineer with experience building, deploying, and productionizing AI/ML, Generative AI, LLM, RAG, and autonomous agent systems across SaaS and PaaS environments. Skilled in developing scalable ML pipelines, model APIs, cloud-based deployments, MLOps/LLMOps workflows, CI/CD automation, model lifecycle management, and enterprise-grade AI solutions. Strong collaborator with data scientists, engineering teams, and stakeholders to transform experimental models into secure, reliable, maintainable, and production-ready machine learning products.

CORE SKILLS

Machine Learning • Deep Learning • Generative AI • Large Language Models • Natural Language Processing • Retrieval-Augmented Generation • MLOps • LLMOps • Model Deployment • Model Serving • ML Pipelines • CI/CD Pipelines • REST APIs • FastAPI • Flask • Python • Scikit-Learn • TensorFlow • Keras • PyTorch • Hugging Face Transformers • OpenCV • Pandas • NumPy • PySpark • MLflow • ZenML • Docker • GitHub Actions • AWS SageMaker • AWS Bedrock • AWS Lambda • AWS EC2 • Google Vertex AI • Google Cloud Run • Google BigQuery • Azure ML • Azure AI • Azure App Services • LangChain • LangGraph • LlamaIndex • AutoGen • LangSmith • Vector Databases • Pinecone • FAISS • Chroma DB • SQL • NoSQL • Model Monitoring • Hyperparameter Tuning • Predictive Modeling • Statistical Analysis • Data Visualization • Agile Methodologies • Microservices Architecture • Stakeholder Management • Team Leadership

PROFESSIONAL EXPERIENCE

Senior Data Scientist, Machine Learning & Generative AI Engineer - Volunteer | NVIDIA

08/2024 - Present

- Contributed to NVIDIA and Z by HP Developer Programs by testing GPU-optimized applications and enterprise-level Generative AI solutions.
- Developed and evaluated open-source and enterprise LLM solutions, including vLLM-based deployments for Generative AI use cases.
- Integrated NVIDIA tools and SDKs to optimize AI applications for performance, scalability, and production readiness.
- Collaborated with NVIDIA and Z by HP teams to validate AI tools, frameworks, and deployment environments.
- Contributed to open-source LLM initiatives supporting innovation in large-scale AI model deployment.

Senior Data Scientist & Machine Learning Engineer | BLOCBELT

12/2022 - 12/2024

- Led development of AI-driven SaaS and PaaS solutions while managing a team of three across data science, AI engineering, big data, and analytics initiatives.
- Delivered AI solutions that improved operational efficiency and generated \$2M in client cost savings.
- Improved sales forecasting accuracy by 80-90% through predictive modeling and analysis across 120+ market entities.
- Launched a personalized fashion e-commerce platform with an advanced recommendation engine, driving an 80% increase in sales and a 200% increase in online orders.
- Developed statistical models, predictive analytics, and dashboards that increased company revenue by 18% and improved data visibility by 45%.

- Integrated AI capabilities into SaaS and PaaS offerings and conducted 25+ A/B tests to optimize digital customer experiences.

AI Data Scientist Kaggle Master Mentor - Part-Time | GOOGLE-KAGGLE

12/2022 - 11/2024

- Mentored aspiring data scientists through the Kaggle-X BIPOC Mentorship Program, dedicating 20 hours weekly to technical guidance and project support.
- Led a team of four mentees in machine learning, data science, problem-solving, and community contribution practices.
- Ranked among the Top 100 Kaggle contributors globally and recognized as the first Pakistani mentor in the program.
- Collaborated on high-impact AI projects and contributed to the global data science and machine learning community.

Data Scientist | Crypto-Express

01/2022 - 12/2022

- Developed an anti-spoofing face recognition application that enhanced digital identity security and reduced identity fraud by 70%.
- Applied machine learning and deep learning techniques to improve operational efficiency by 25% and reduce costs by 15%.
- Built predictive models and delivered actionable insights that contributed to a 20% revenue increase.
- Performed exploratory data analysis and machine learning experimentation, saving 250+ hours of manual analysis annually.
- Deployed AI/ML projects remotely while adapting solutions to dynamic business and technical requirements.

Artificial Intelligence Engineer | Pakistan Freelancing Training Center

01/2021 - 12/2021

- Led AI and data science training sessions focused on practical machine learning applications and real-world problem solving.
- Designed and deployed machine learning models using TensorFlow and Keras to improve system efficiency and accuracy.
- Automated repetitive data tasks using Python, reducing manual effort and minimizing error rates.
- Conducted AI research and translated emerging methodologies into practical project implementations.

PROJECTS

AutoML-Studio

Built a low-code/no-code machine learning platform for rapid predictive model development and deployment. Integrated MLflow for experiment tracking, SHAP for explainability, data drift detection for reliability, and FastAPI and Streamlit for serving classification, regression, time-series forecasting, and clustering workflows.

Customer Satisfaction Prediction System

Created an MLOps-based predictive analytics system using ZenML and MLflow to forecast customer satisfaction with 93% accuracy. Streamlined pipeline management, model tracking, and deployment to support data-driven retention decisions.

Conversational AI Chatbot for Customer Support

Designed and deployed an enterprise-grade GPT-4o chatbot integrated with LangChain and AWS Bedrock. Implemented intent recognition and dynamic response generation, reducing customer query resolution time by 40%.

Health Assistance Application

Architected a healthcare-focused LLM application using Gemini 1.5 Pro, RAG, and LangChain to support medical information retrieval, symptom guidance, treatment suggestions, and preventive healthcare recommendations.

DocuWiz AI

Engineered a document intelligence solution using BART, DistilBERT, RAG, and LLMs for text extraction, summarization, sentiment analysis, and PDF/DOCX insights. Deployed with Streamlit and FastAPI, reducing document processing time by 60%.

AI-Powered Blog Generator

Developed an AI content generation platform using Llama 3.1 8B, AWS Bedrock, RAG, and Streamlit to automate blog creation, editing, and publishing for enterprise content workflows.

Harvestify AI-Powered Plant Health Assistant

Designed an Azure-based computer vision application using Azure AI Studio and Azure App Services to diagnose plant leaf diseases from images, recommend crops and fertilizers, reduce disease detection time by 70%, and improve crop yield predictions by 30%.

Intelligent Document Summarization Tool

Developed a transformer-based summarization system using Pegasus to generate concise summaries from PDF and DOCX documents. Deployed with FastAPI and Streamlit, reducing manual review time by 60%.

EDUCATION

Bachelor in Computer Science, Major: Artificial Intelligence and Computer Science | Institute of Management Sciences

CERTIFICATIONS

- IBM Machine Learning Specialization Professional Certificate - IBM
- Deep Learning Specialization - DeepLearning.ai
- Microsoft Azure AI Fundamentals AI-900 Exam Prep Specialization - Microsoft
- Prompt Engineering for ChatGPT - Vanderbilt University
- Machine Learning Engineering for Production (MLOps) - DeepLearning.ai
- AI Product Management - Duke University
- Generative Adversarial Networks (GANs) Specialization - DeepLearning.ai
- IBM Generative AI Product Managers - IBM
- Preparing for Google Cloud Certification: Machine Learning Engineer - Google
- IBM AI Product Manager - IBM
- AWS Cloud Solutions Architect Professional Certificate - Amazon Web Services
- Google Project Management: Professional Certificate - Google
- Practical Data Science on the AWS Cloud Specialization - DeepLearning.ai
- Machine Learning Specialization - DeepLearning.AI
- MLOps | Machine Learning Operations Specialization - Duke University
- Google Business Intelligence Professional Certificate - Google
- Advanced Machine Learning on Google Cloud Specialization - Google Cloud
- Large Language Model Operations (LLMOps) - Duke University
- Google Advanced Data Analytics Professional Certificate - Google

ACHIEVEMENTS

- Top 100 Global Kaggle Master
- KaggleX BIPOC Mentor
- 24x LinkedIn Top Voice
- Contributor to NVIDIA Developer Program
- Contributor to Google Developer Group
- Contributor to AWS AI Community
- Delivered enterprise-grade AI solutions across Generative AI, RAG, LLMOps, MLOps, SaaS, and PaaS environments
- Generated \$2M in client cost savings through AI-driven SaaS and PaaS solutions
- Improved sales forecasting accuracy by 80-90% across 120+ market entities